

Lecture 3

randomized experiments and Fisher's exact p-value

Outline

- Randomized experiment mechanisms
- Fisher's exact p-value
 - Fisher's original experiment
 - Hypothesis testing
 - Construct confidence intervals
 - Choice of the test statistics
- Suggested reading: Imbens and Rubin Chapter Chapter 4-5, Peng's book
Chapter Chapter 3

Potential outcomes and observed outcomes

For each unit i , define $Y_i(0)$, $Y_i(1)$, and baseline covariates X_i .

$$W_i \in \{0,1\}, \quad Y_i^{obs} = W_i Y_i(1) + (1 - W_i) Y_i(0)$$

Well-defined treatments and no interference support this notation.

Hold the study units and all potential outcomes fixed.

Random assignment determines which outcome we observe.

Randomization still leaves one potential outcome missing for each unit.

Treatment assignment mechanism

- **Assignment vector** for binary treatment with N units: $\mathbf{W} = (W_1, \dots, W_N) \in \{0,1\}^N$
- The design specifies a probability for every assignment vector w .

- **Unconfoundedness property:**

$$P(W = w \mid X, Y(0), Y(1)) = P(W = w \mid X)$$

- Assignment mechanism does not depend unobserved $\mathbf{U}$ pretreatment confounders
- $\mathbf{U}$ includes potential outcomes $(Y_i(0), Y_i(1))$
- We can alternatively understand it as

$$W_i \perp (Y_i(0), Y_i(1)) \mid \mathbf{X}_i$$

- Make the treatment and control groups “identical”

$$P(Y_i(0), Y_i(1) \mid \mathbf{X}_i, W_i = 0) = P(Y_i(0), Y_i(1) \mid \mathbf{X}_i, W_i = 1)$$

- Both treatments are possible for each unit. Realized groups can still differ. **Propensity score:**

$$0 < e_i = P(W_i = 1 \mid X) < 1$$

Common designs of randomized experiments

- Five examples of randomized experiment mechanisms
 - Bernoulli trial
 - Completely randomized experiment
 - Stratified randomized experiment
 - Paired randomized experiment
 - Rerandomization
- The purpose of restricting the assignment mechanism is to eliminate assignment vectors that are less desirable for estimating causal effects
 - Examples: all males get treatment; all females get control

Bernoulli randomization

Independently assign treatment with known probability e_i .

$$W_i \mid X \sim \text{Bernoulli}(e_i), \quad i = 1, \dots, N$$

$$P(W = w \mid X) = \prod_{i=1}^N e_i^{w_i} (1 - e_i)^{1 - w_i}$$

$$\text{Fair coins: } e_i = 1/2, \quad P(W = w) = 2^{-N}$$

With four units, what is the probability that everyone gets the same treatment?

Complete randomization

Choose exactly N_1 of the N units to receive treatment.

$$P(W = w) = \begin{cases} \binom{N}{N_1}^{-1} & \text{if } \sum_{i=1}^N w_i = N_1 \\ 0 & \text{otherwise} \end{cases}$$

The group sizes are fixed, and the W_i are dependent (slightly negatively associated).

Two women and two men, with two treated:
could both treated units be men?

Stratified randomization

Block (strata) j contains N_j units, with exactly N_{1j} assigned to treatment.

Completely randomize within each block, independently across blocks.

$$P(W = w \mid X) = \begin{cases} \prod_{j=1}^J \binom{N_j}{N_{1j}}^{-1} & \text{if } \sum_{i: B_i=j} w_i = N_{1j} \text{ for all } j \\ 0 & \text{otherwise} \end{cases}$$

Blocking on predictors of the outcome can improve precision.

For two women and two men, treat one person of each sex.

Pair work: count and assign probabilities

Four units: women A and B, men C and D

Design	Number allowed	P(one allowed w)
Independent fair coins	?	?
Exactly two treated	?	?
One treated within each sex	?	?

Which designs allow $w = (1, 1, 0, 0)$?

Assignment counts and probabilities

Design	Number allowed	P(one allowed w)
Independent fair coins	16	1/16
Exactly two treated	6	1/6
One treated within each sex	4	1/4

Complete randomization: AB, AC, AD, BC, BD, CD

Stratified randomization: AC, AD, BC, BD

Stratification in the Moderna trial

- Participants received vaccine or placebo in a 1:1 ratio, using a centralized interactive response technology system
- Randomization used three baseline risk groups:
 - Age 65 or older
 - Under 65, at heightened risk of severe COVID-19
 - Under 65, without heightened risk

Why are age and baseline risk useful blocking variables?

Paired randomization

Match $N = 2J$ units into J pairs using baseline covariates.

$$P(W = w \mid X) = \begin{cases} 2^{-J} & \text{if } \sum_{i \text{ in pair } j} w_i = 1 \text{ for all } j \\ 0 & \text{otherwise} \end{cases}$$

Randomly treat one unit per pair. There are 2^J allowed assignments.

Mexico's Seguro Popular evaluation used 74 matched pairs of health-facility catchment areas.

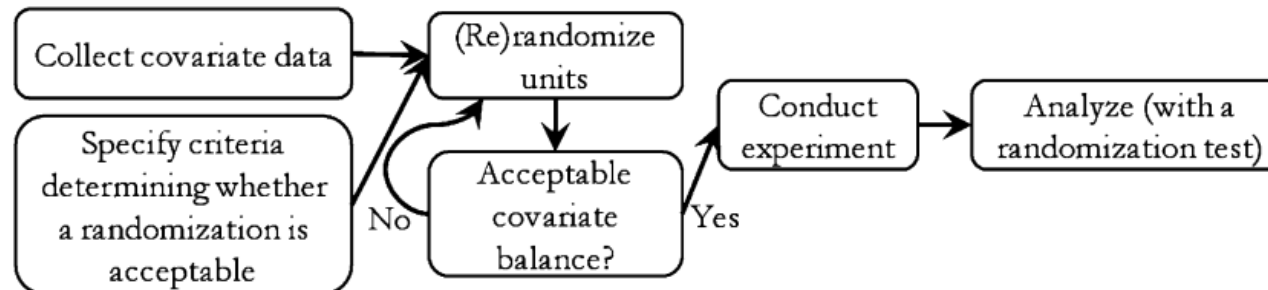
Rerandomization

- Prespecify a baseline-balance rule $A(w, X) \in \{0,1\}$.
- Draw from the base design $p_0(w | X)$. Accept only when $A = 1$.

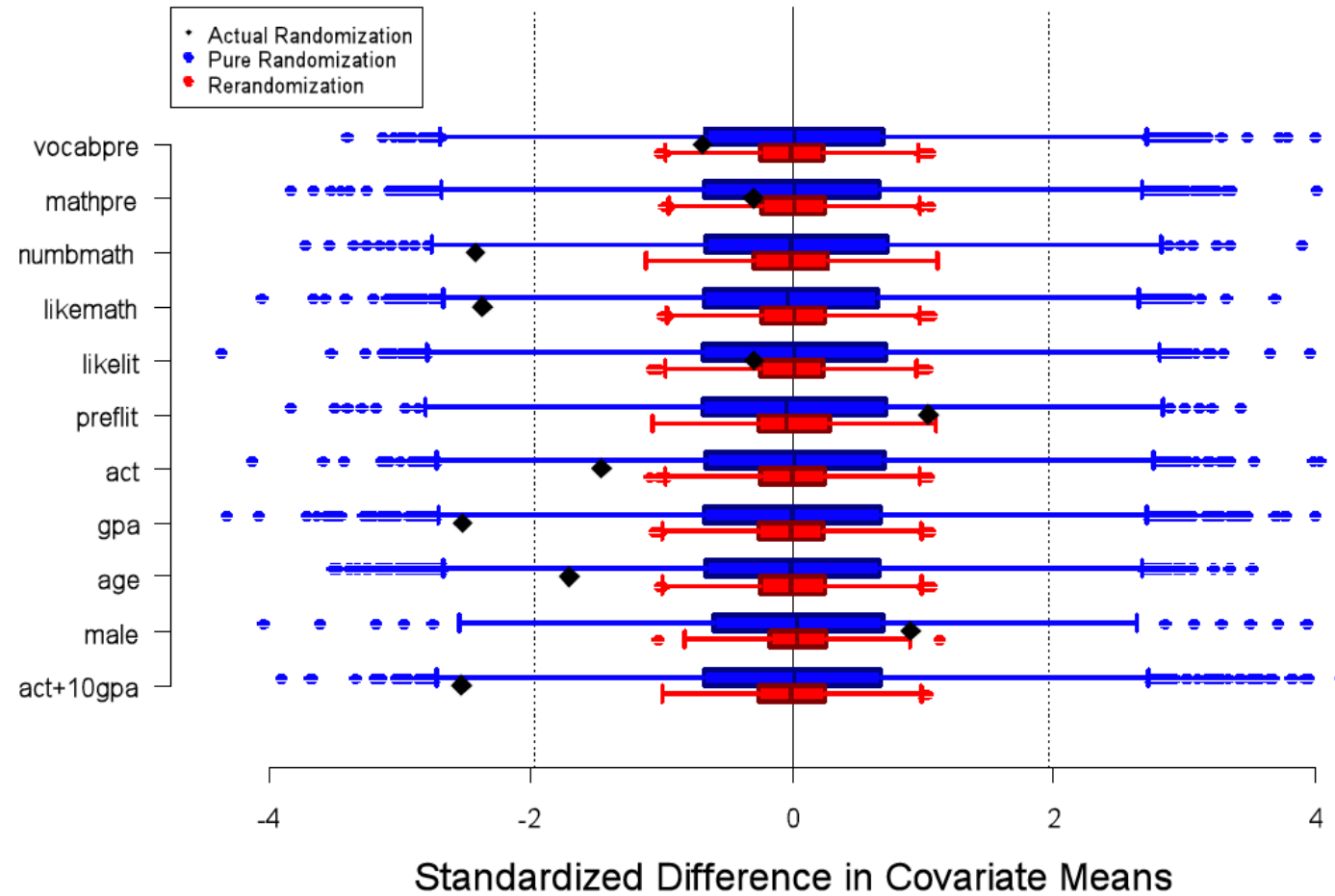
$$P(W = w | X, A = 1) = \frac{p_0(w | X) A(w, X)}{\sum_v p_0(v | X) A(v, X)}$$

The restriction removes unacceptable assignments from the support

- Improve covariance balancing for the realized experiment



Rerandomization: baseline balance



Randomization Inference vs. Model-based Inference

- Randomization as the “**reason basis for inference**” (Fisher)
- Randomness comes from the physical act of randomization, which then can be used to make statistical inference
- Also called **design-based inference**
- Advantage: design justifies analysis
- model-based inference: assume a distribution for potential outcomes (at least the i.i.d. assumptions)
- Advantage of model-based inference: flexibility
- Two types of classical randomization inference
 - Fisher’s exact p-values
 - Neyman’s repeated sampling approach

Fisher's sharp null

$$H_0: Y_i(1) = Y_i(0) \text{ for every } i$$

Under this null, every missing potential outcome is known:

$$Y_i(0) = Y_i(1) = Y_i^{obs}$$

Reassign treatment according to the design and recompute the statistic.

The study units and outcomes stay fixed under H_0 . The assignment W changes.

Fisher's original experiment: Lady tasting tea

[Fisher, 1935]

- The lady in question (Muriel Bristol) claimed to be able to tell whether the tea or the milk was added first to a cup.
- Fisher proposed to give her **eight cups, four of each variety**, in random order.
- **Null hypothesis:** the lady cannot tell the difference
- **How to define the causal effect?**
 - whether the tea or the milk was added first has any effect on the lady's guess result
- **What is a unit?**
- **What is the treatment assignment?**
 - $W_i = 1$ if tea is added first
- **What is $Y_i(0)$ and $Y_i(1)$?**
 - The lady's potential guess results
- **Sharp null:** $H_0: Y_i(0) \equiv Y_i(1)$ for every cup
 - Under H_0 , her response labels stay fixed across assignments



Tea: assignments and correct classifications

$$Y_i(0) = Y_i(1) = Y_i$$

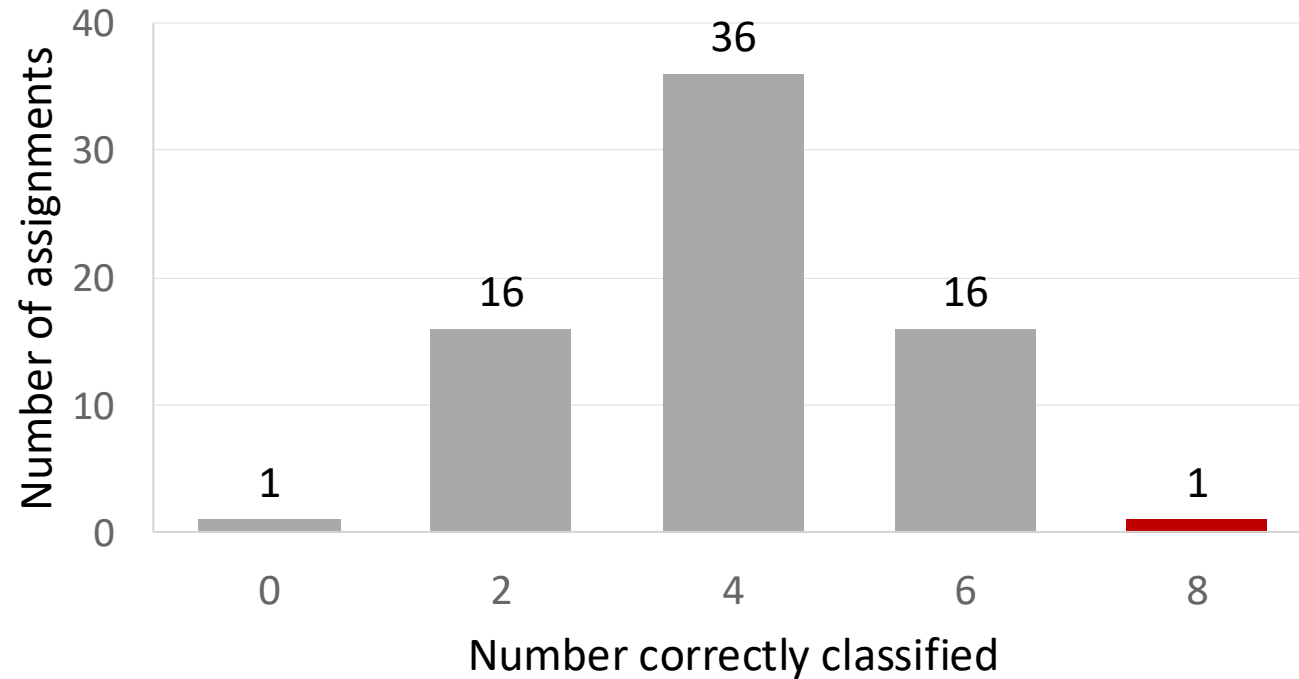
W_i

Cup	1	2	3	4	5	6	7	8	Correct
Fixed labels (guess)	M	T	T	M	M	T	T	M	
Assignment 1 (actual)	M	T	T	M	M	T	T	M	8
Assignment 2	T	T	M	M	T	T	M	M	4
Assignment 3	T	M	M	T	T	M	M	T	0

How many preparation assignments are allowed?

How many give perfect classification? Why is 2^8 the wrong denominator?

The tea randomization distribution



Possible assignments:

$$\binom{8}{4} = 70$$

Observed $K = 8$

$$p = P(K \geq 8 \mid H_0) = \frac{1}{70} \approx 0.014$$

Pair work: cough frequency

Unit	W_i	$Y_i(0)$	$Y_i(1)$	Y_i observed
1	1	?	3	3
2	1	?	5	5
3	1	?	0	0
4	0	4	?	4
5	0	0	?	0
6	0	1	?	1

Under sharp null H_0 , fill in the missing outcomes and calculate the observed difference in means:
 $D = \text{treated mean} - \text{control mean}$.

Then suppose the treat units are $\{1, 2, 4\}$. What is the new D under the hypothetical assignment?

Cough: imputation and reassignment

Unit	$Y_i(0)$	$Y_i(1)$
1	3	3
2	5	5
3	0	0
4	4	4
5	0	0
6	1	1

Observed treated set {1, 2, 3}

$$D = \frac{8}{3} - \frac{5}{3} = 1$$

New treated set {1, 2, 4}

$$D = \frac{12}{3} - \frac{1}{3} = \frac{11}{3}$$

Under H_0 , the outcomes stay fixed and the assignment determines D .

Cough: the exact p-value

$$D = \bar{Y}_1 - \bar{Y}_0, \quad T = |D|, \quad T^{obs} = 1$$

 D 	1/3	1	5/3	7/3	11/3
Number of assignments	4	6	6	2	2

$$p = \frac{6 + 6 + 2 + 2}{20} = 0.80$$

Minimum two-sided $p \geq 2/20 = 0.10$. A 5% rejection is impossible.

- $N = 6$ is too small to obtain statistically significant rejections

Fisher's exact p-value

Let $\mathcal{W}$ contain the assignments allowed by the design.

$$p = \sum_{w \in \mathcal{W}} P(W = w \mid X) 1\{T(W) \geq T^{obs}\}$$

If every allowed assignment has the same probability:

$$p = \frac{\#\{w \in \mathcal{W}: T(W) \geq T^{obs}\}}{|\mathcal{W}|}$$

A sharp null supplies the outcomes. The design supplies the probabilities.
The justification is by design alone.

The reference distribution follows the design

Design	Reference assignments
Bernoulli	Draw independent W_i with the specified probabilities e_i
Complete	Choose exactly N_1 treated units
Stratified / paired	Preserve the treated count within every block or pair
Rerandomization	Apply the original baseline-balance acceptance rule

A paired experiment is analyzed by shuffling labels across all units.
Which assignments does this incorrectly allow?

Sharp null and zero average effect

$$H_0: Y_i(1) - Y_i(0) = 0 \text{ for every } i$$

This sharp null determines every missing potential outcome.

$$\text{Zero average effect null } H_0: \frac{1}{N} \sum_{i=1}^N [Y_i(1) - Y_i(0)] = 0$$

- Effects +2 and -2 satisfy the average null, but violate the sharp null.

One average restriction is insufficient to impute all missing outcomes.

A common additive effect

$$H_0(\tau_0): Y_i(1) - Y_i(0) = \tau_0 \text{ for every } i$$

Each candidate τ_0 supplies a complete potential-outcome schedule:

$$Y_i(0) = Y_i^{obs} - W_i^{obs} \tau_0$$

$$Y_i(1) = Y_i(0) + \tau_0$$

For a new assignment w , the observed outcome becomes $Y_i(0) + w_i \tau_0$.

Worked example: a common effect of 1

Eight units, with exactly five treated. Test $H_0(1): Y_i(1) - Y_i(0) = 1$.

Unit	1	2	3	4	5	6	7	8
W_i observed	1	1	0	1	1	0	0	1
Y_i observed	12	7	2	7	6	6	2	7
$Y_i(0)$	11	6	2	6	5	6	2	6
$Y_i(1)$	12	7	3	7	6	7	3	7

Treated unit 1: $Y_1(0) = 12 - 1 = 11$.

Control unit 3: $Y_3(1) = 2 + 1 = 3$.

The statistic under a candidate effect

$$T_{\tau_0}(w) = |D_{\tau_0}(w) - \tau_0|$$

Observed assignment:

$$D^{obs} = \frac{39}{5} - \frac{10}{3} = \frac{67}{15}, \quad T^{obs} = \left| \frac{67}{15} - 1 \right| = \frac{52}{15}$$

New treated set {1, 2, 3, 4, 5}: $T = |30/5 - 14/3| = 4/3$.

$$p(1) = \frac{11}{56} \approx 0.196$$

Enumerate all 56 assignments and include statistics tied with T observed.

Confidence intervals by test inversion

$$Y_i(1) - Y_i(0) = \tau \text{ for every } i$$

For each candidate τ_0 , impute its outcomes and compute $p(\tau_0)$.

$$C_{1-\alpha} = \{\tau_0 : p(\tau_0) > \alpha\}$$

At $\alpha = 0.05$, retain every candidate that the test does not reject.

The confidence set concerns a common additive effect.
When the retained set is an interval, report its endpoints.

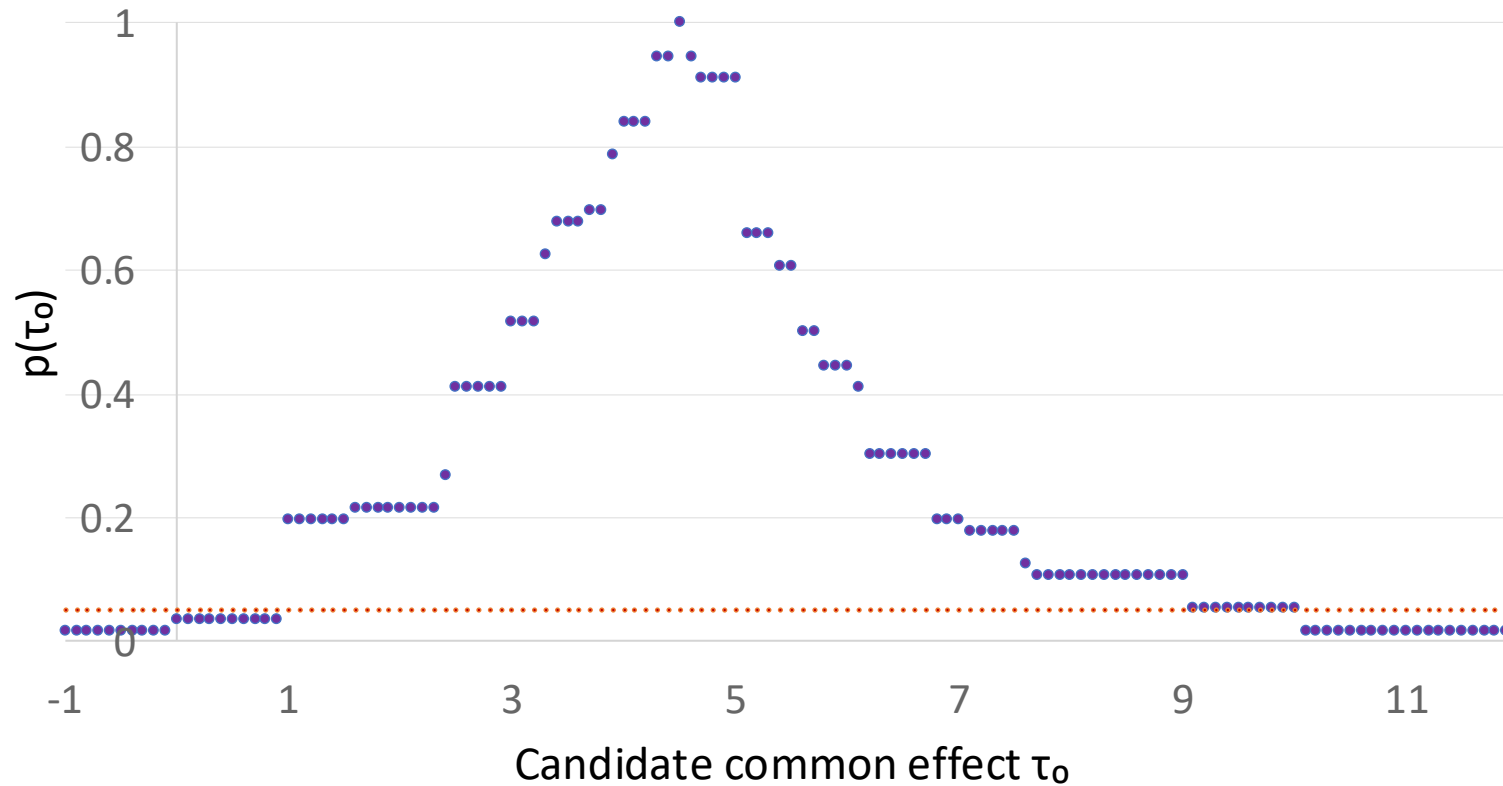
Pair check: which effects remain plausible?

Same eight units. Reject $H_0(\tau_0)$ when $p(\tau_0) \leq 0.05$.

Candidate τ_0	Extreme assignments	Exact p-value
0	2 / 56	0.0357
0.9	2 / 56	0.0357
1	11 / 56	0.1964
4	47 / 56	0.8393
8	6 / 56	0.1071
10	3 / 56	0.0536
10.1	1 / 56	0.0179

Which candidates belong to the 95% confidence set?

The inverted confidence interval



95% confidence set

$[1, 10]$

Both endpoints
are retained.

Points evaluate τ_0 in steps of 0.1. The interval boundaries were checked exactly.

Zero is excluded under the common additive effect model.

Monte Carlo approximation

- Exact computation is difficult when N is large
- When enumeration is expensive, randomly draw B assignments from the design.

$$b = \sum_{r=1}^B 1\{T^{(r)} \geq T^{obs}\}, \quad \hat{p}_{MC} = \frac{b + 1}{B + 1}$$

- Monte Carlo approximation
 1. Impute outcomes under the candidate sharp null.
 2. Draw assignments using the original probability law.
 3. Recompute the statistic and count extreme results.

Increasing B improves simulation accuracy. It does not increase N .

Simulation accuracy with $N = 72$

Number of possible assignments: $\binom{72}{36} \approx 4.43 \times 10^{20}$

P-value computation with Monte Carlo approximation

Number of Simulations	P-Value	$\widehat{\text{(s. e.)}}$
100	0.010	(0.010)
1,000	0.044	(0.006)
10,000	0.044	(0.002)
100,000	0.042	(0.001)
1,000,000	0.043	(0.000)

Note: Statistic is absolute value of difference in average ranks of treated and control cough frequencies. P-value is proportion of draws at least as large as observed statistic.

What changes when B increases? What remains fixed?

Choice of test statistic

Cough unit	1	2	3	4	5	6
Outcome	3	5	0	4	0	1
Rank (average ties)	4	6	1.5	5	1.5	3

$$T_{mean} = |\bar{Y}_1 - \bar{Y}_0 - 0|, \quad T_{rank} = |\bar{R}_1 - \bar{R}_0|$$

- Both give $p = 0.80$ here. Ranks reduce sensitivity to extreme magnitudes.

Choose the statistic and direction before examining the results.

Ranks under a nonzero common effect

- We define the rank of each unit based on $\text{rank}(Y_i(0))$ [or equivalently $\text{rank}(Y_i(1))$], instead of $\text{rank}(Y_i^{\text{obs}})$

$$Y_i(0) = Y_i^{\text{obs}} - W_i^{\text{obs}} \tau_0$$

$$R_i(\tau_0) = \text{rank}(Y_i(0))$$

$$T_{\tau_0}(w) = |\bar{R}_{\text{treated}} - \bar{R}_{\text{control}}|$$

Recompute adjusted outcomes and their ranks for each candidate τ_0 .

Illustration of Fisher's randomization test

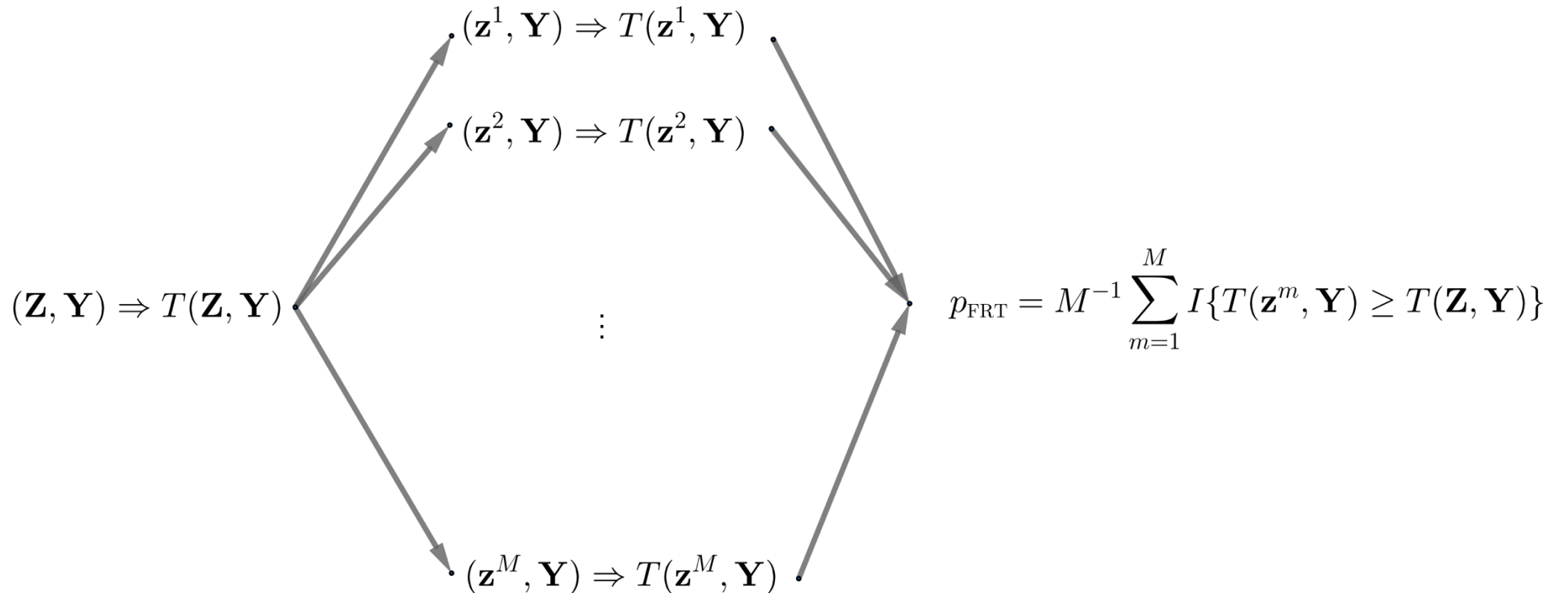


Figure 3.1 of Peng's book

What determines a randomization test?

The design determines allowed assignments and their probabilities.

The sharp null supplies the missing potential outcomes.

The statistic defines which results count as extreme.

Why did the cough example have $p = 0.80$?

What assumption supports the eight-unit confidence interval $[1, 10]$?